

Focus Guard AI: Technical Project Report

1. Executive Summary

Focus Guard is an intelligent, emotion-aware desktop assistant designed to enhance productivity. Unlike a passive speaker, it uses real-time Computer Vision to monitor the user's facial expressions and dynamically adjusts the auditory environment. If stress or distraction is detected, it automatically switches to calming music (Rescue Protocol). If focus is maintained, it plays productivity tracks. The system runs locally on CPU/GPU (Metal) and integrates seamlessly with Spotify via OAuth 2.0.

2. Technical Architecture

Hardware Stack:

- Compute: MacBook Pro (M5 Chip) with Metal Acceleration.
- Input: Integrated Webcam (cv2.VideoCapture).
- Output: System Audio / Bluetooth Speaker.

Software Stack:

- Language: Python 3.11
- AI/ML: TensorFlow 2.15 (Metal Plugin), Keras
- Computer Vision: OpenCV (Haar Cascades)
- Audio: Spotipy (Spotify Web API), Pygame (Local Fallback)
- Environment: Virtualenv, Dotenv for Security

3. Methodology: The AI Core

3.1 Model Selection: Mini-Xception

We chose Mini-Xception (CNN) over heavy architectures like ResNet50 for real-time edge performance. It utilizes Depthwise Separable Convolutions to reduce parameters by ~90% while maintaining high accuracy on the FER-2013 dataset (Facial Expression Recognition).

3.2 Preprocessing Pipeline

- Grayscale Conversion: Reduces dimensionality (RGB -> 1 channel).
- Histogram Equalization: Normalizes lighting conditions (contrast enhancement).
- Resizing: Standardized input to 48x48 pixels.
- Normalization: Scaled pixel values to [0, 1] range.

3.3 Stabilization (Anti-Jitter)

To prevent bounding box flickering, we implemented an Exponential Moving Average (Low-Pass Filter):

$$\text{New_Pos} = (\text{Current} * \alpha) + (\text{Old_Pos} * (1 - \alpha))$$

We tuned $\alpha=0.5$ for instant responsiveness.

4. Application Logic & State Machine

The system operates as a State Machine with four primary states:

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1. FOCUS (Neutral): Plays Deep Work / Hollywood Soundtracks.
2. STRESS (Negative Emotions): Triggers 'Rescue Protocol' (Bollywood/Rain).
3. HAPPY (Positive): Plays Gujarati Romantic Hits (Locked for 90s).
4. DISTRACTION (Surprise): Pauses Audio.

Key Algorithms:

- Argmax Classification: We use 'Dominant Emotion' logic rather than arbitrary thresholds. If 'Neutral' is the highest probability, the system prioritizes Focus.
- Hysteresis (Sticky States): A 5-second sustenance timer prevents rapid switching. Users must hold an expression for 5s to trigger a music change.
- Dynamic Cooldown: Happy mode enforces a 90s lock-in to prolong positive vibes.

5. Integration Details

- Spotify API: Automated OAuth 2.0 flow. Handles token refreshing and device handoff.
- Volume Control Bypass: Automatically detects if the target device supports volume fading. If restricted (e.g., Mobile/Web), it skips the fade animation to prevent API errors.
- Offline Fallback: If Spotify fails or internet drops, the system falls back to local assets (rain.wav, lofi.wav).

6. Conclusion

Focus Guard successfully demonstrates a complete end-to-end ML pipeline. It combines lightweight Deep Learning with robust software engineering principles (State Machines, API Integration, Error Handling) to create a product that feels 'alive'. Future work involves porting to Raspberry Pi 4 via TFLite for a fully headless experience.